Road Marking Paint

PHOSPHORESCENT NANO LATEX BEADS

GROUP 11

PH 2006/2022/ Group Number-11

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Each member's contribution to the project:

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Abstract

Road marking visibility has made a drastic improvement during the past few years. The most common solution for this is using retroreflective beads. While they resolve the matter, they don't provide complete efficiency under some conditions. As an alternative to this, phosphorescent nano latex beads are introduced. With the use of the solvent evaporation method phosphorescent dye made from strontium alumina is injected into the nano latex beads, which will then be able to absorb the solar energy during the daytime and emit at night as green light. The report provides an in-depth explanation of the scientific processes involved in the creation of phosphorescent nano latex.

Keywords

Retroreflectivity, glass beads, latex beads, phosphorescent

Introduction

Road marking provides guidance to drivers and pedestrians and ensures their safety. Paints are commonly used for road marking, and they are composed of 3 fundamental components: pigments, resins or binders, and water. Pigments are finely ground powders that are responsible for achieving the desired color intensity within the paint. Resins or binders serve as the adhesive agent within the paint, effectively binding pigments, and glass beads to the road surface. Pigment and resins are mixed with water.

Various types of road marking paints exist, including Hot Thermoplastic, MMA, Solvent-Based Acrylic, Water-Based, Epoxy, and Pre-formed Thermoplastic Tape. Among these, Hot Thermoplastic paint stands as one of the most widely preferred choices because of its durability and quick drying time. Additives are added to the paint to enhance the plasticity of the coating ensuring better adhesion to the road surface, while glass beads work by reflecting light from vehicle headlights preferentially back toward the light source (retro reflectance) allowing road markings to be visible at night.

The drawbacks of using glass beads: (1) In places where a thick layer of paint should be applied, due to the high density of glass beads it tends to be deposited at the bottom. (2) During heavy rains the water covers the glass beads. In both cases, it causes a decrease in the retro-reflective property of the paint. For the first drawback, latex beads offer a good solution, because of their lower density property than glass beads. The main issue is that latex beads have less reflective properties than glass beads.

To address this issue, this case study focuses on creating phosphorescent nano latex beads to enhance retro-reflection.

photoluminescent pigments are specialized compounds that have the ability to absorb and store ambient light during the daytime and subsequently release it as a visible and often radiant glow during the night. Photoluminescent materials can be divided into two categories based on their luminescence duration, fluorescence and phosphorescence. Both Phosphorescent and fluorescent materials have the unique property of emitting a visible glow for several seconds to even a couple of days after they've absorbed solar energy. But Phosphorescent materials have a higher luminescent duration than fluorescent materials. Therefore, using phosphorescent material is more effective than using fluorescent material. In the past, most applications of phosphorescence utilized zinc sulfide. But it has weak luminescent capabilities compared to the strontium aluminate. Therefore, making phosphorescent dye with strontium aluminate can increase the luminescence of the dye.

Instead of mixing that phosphorescent dye directly into the road marking paint, adding that phosphorescent dye with strontium aluminate injected inside the nano latex beads will affect the longevity of the luminous effect. Also, it emits green light, making it more visible to the human eye.

The solvent evaporation method is commonly used to make polymeric nanoparticles. This method is used for drug delivery, where the drug is encapsulated by polymeric nanoparticle coating. In order to produce latex nanoparticles with phosphorescent dye encapsulated in them, it follows the same steps. With this, we can enhance the retro-reflective property while ensuring the safety of nighttime drivers.

Solvent evaporation method to create phosphorescent nano latex beads.

Initially, the latex polymer is dissolved in a volatile organic solvent like dichloromethane, chloroform, and ethyl acetate. And within this solution, the phosphorescent dye is also dissolved. Then, the resulting solution is added to the aqueous phase, which already contains a surfactant, and subjected to highintensity homogenization to create an emulsion. Emulsion can be formed using the two conventional methods; (1) single emulsions, and (2) double emulsions. After achieving a stable emulsion, the organic solvent can be removed by either raising the temperature under reduced pressure or through continuous stirring, resulting in the dispersion of nanodroplets.

As the organic solvent is gradually removed through evaporation, the phosphorescent dye and latex polymer within the solution precipitate within the droplets, resulting in the formation of latex nanospheres or nano capsules. The size of the nano latex particles can be controlled by manipulating parameters such as adjusting the evaporation temperature, controlling the evaporation rate, and varying the stirring speed during the process. Finally, the nano latex beads produced have a core made of nanoparticles containing phosphorescent dye enclosed within a latex polymer shell.

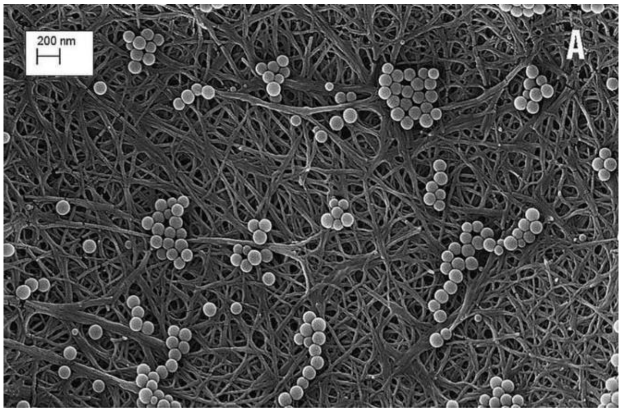

Figure 1: Nano latex beads before injecting phosphorescent dye

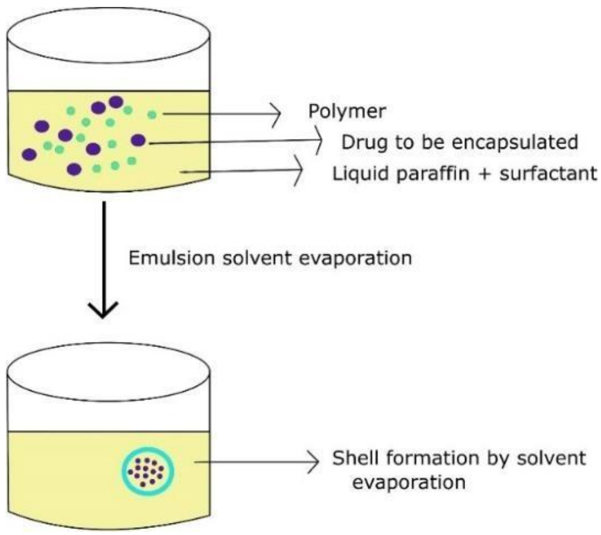

Figure 2: solvent evaporation method to inject phosphorescent dye into nano latex beads.

Market analysis

Market Size and Growth:

The global traffic road marking coatings market exhibited robust growth, with a valuation of USD 4.92 billion in 2021. It is poised for continued expansion, projected to maintain a compound annual growth rate (CAGR) of 6.0% from 2022 to 2030. This growth reflects the escalating demand for advanced road marking solutions worldwide. Additionally, the global road marking paints & and coatings market is expected to witness substantial growth, estimated to increase from US$ 1,902.0 million in 2023 to US$ 2,899.8 million by 2033. These figures underscore the promising opportunities in the road marking industry.

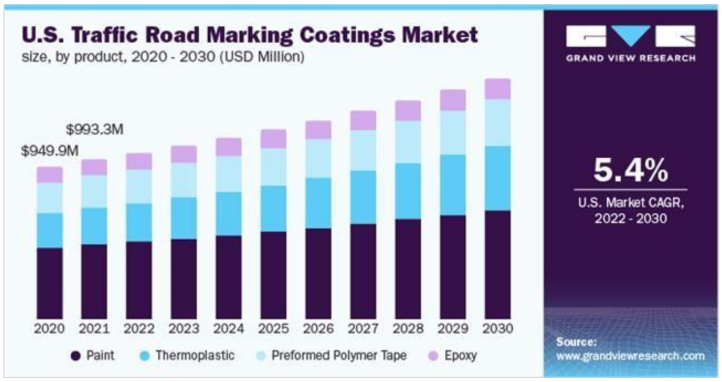

Figure 3 : U.S. Traffic road marking coating's market analysis

Target Customer Base:

Our primary customers will include government entities such as the Road Development Authorities (RDAs), construction companies engaged in infrastructure projects, as well as private entities responsible for managing parking lots and private roads. Catering to these diverse customer segments allows us to tap into various facets of the roadmarking market and align our product offerings with their specific needs.

Innovative Solution:

Our revolutionary concept involves replacing traditional glass beads in road marking paint with phosphorescent nano latex beads. This innovation offers a multifaceted advantage. Firstly, it significantly enhances the eco-friendliness of our product, aligning with global sustainability trends and regulatory requirements. Secondly, the inclusion of phosphorescent dyes imparts a captivating nighttime glow to road markings, substantially improving visibility and potentially reducing nighttime accident rates. This innovation positions our product as a beacon of safety and environmental consciousness within the industry.

Distribution Channels:

To ensure widespread availability, we plan to employ a comprehensive distribution strategy. This includes direct sales channels for engaging with government agencies and private entities directly. Additionally, we will partner with retailers to reach a broader audience and explore online platforms to facilitate convenient access for customers.

Marketing Strategy:

Our marketing strategy encompasses a multifaceted approach. We will leverage advertising campaigns to create awareness about our innovative road marking solution. Public relations efforts will focus on building trust and credibility in the industry. Active engagement on social media platforms will keep our audience informed and engaged, while online marketing strategies will maximize our online presence. By employing these methods, we aim to establish our brand as a pioneer in the field of road marking coatings.

Conclusion

The innovative approach of replacing traditional glass beads with phosphorescent nano latex beads presents a transformative solution for enhancing road safety and environmental sustainability. The process of making nano latex beads and employing the solvent evaporation method to infuse them with phosphorescent dye enables these beads to absorb sunlight during the day and emit a gentle, radiant glow during the night. Most importantly properties of thermoplastic paint are consistent after replacing glass beads with latex beads since other components remain constant.

Providing a consistent, luminous guidance system can significantly reduce accident rates at night, making roads safer for all users. Furthermore, this enhanced visibility extends to challenging weather conditions, such as high humidity and rainy days, making our road markings more effective even in adverse weather, thereby contributing to accident prevention.

In addition to its safety benefits, our solution also addresses sustainability concerns. By opting for nano latex beads over traditional glass beads, we prioritize eco-friendliness in road marking materials. This choice not only reduces environmental impact but also signifies a commitment to responsible resource utilization.

Moreover, our innovation has the potential to stimulate economic growth. The production and integration of nano latex beads into road marking systems can create new employment opportunities within the latex bead manufacturing industry. As the demand for these eco-friendly alternatives increases, so does the need for skilled workers, thus bolstering the job market and supporting local economies. This technology has multifaceted benefits that have the potential to revolutionize road marking systems.

Figure 4: road marking using phosphorescent paint

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image links

figure1:**https://www.researchgate.net/figure/SEM-images-of-A-100-nm-latex-beads-and-B-50-nmgold-nanoparticles-retained-on-the_fig3_342499957**

figure2:**https://www.researchgate.net/profile/Sobia-Noreen-2/publication/353087984/figure/fig3/AS:1078427771777028@1634128625911/Solvent-evaporationmethod-of-preparation-for-nanoparticles.png**

figure3:**https://www.grandviewresearch.com/static/img/research/us-traffic-road-marking-coatingsmarket.png**

figure4:**https://www.google.com/url?sa=i&url=https%3A%2F%2Fauto.hindustantimes.com%2Fauto%2Fnews%2Fthis-country-to-install-safety-road-markings-that-glow-in-the-dark-41653816712562.html&psig=AOvVaw0E9Vp700-e4UmVhL9Au4lD&ust=1694857273111000&source=images&cd=vfe&opi=89978449&ved=0CBAQjRxqF woTCIijsqeqrlEDFQAAAAAdAAAAABAO**

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Appendix




Memories of our First meeting with our mentor and our group members visiting the field site.